

University Interscholastic League
2023 – 2024 Elementary Number Sense Test B

Contestant's Number _____

Final		
2 nd		
1 st		
	Score	Initials

**Read Directions Carefully
Before Beginning Test**

**Do Not Unfold This Sheet
Until Told to Begin**

Directions: Do not turn this page until the person conducting this test gives the signal to begin. This is a ten-minute test. There are 80 problems. Solve accurately and quickly as many as you can in the order in which they appear. **ALL PROBLEMS ARE TO BE SOLVED MENTALLY.** Make no calculations with paper and pencil. Write only the answer in the space provided at the end of each problem. Problems marked with a (*) require approximate integral answers; any answer to a starred problem that is within five percent of the exact answer will be scored correct; all other problems require exact answers.

The person conducting this contest should explain these directions to the contestants.

Stop – Wait for Signal!

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| <p>(1) $16 + 32 =$ _____</p> <p>(2) $79 + 29 =$ _____</p> <p>(3) $85 - 23 =$ _____</p> <p>(4) $336 \div 3 =$ _____</p> <p>(5) $17 \times 5 =$ _____</p> <p>(6) $614 - 47 =$ _____</p> <p>(7) $8 + 15 + 22 =$ _____</p> <p>(8) $11 \times 27 =$ _____</p> <p>(9) $720 \div 9 =$ _____</p> <p>*(10) $2304 + 699 =$ _____</p> <p>(11) $17 \times 25 =$ _____</p> <p>(12) Which digit is in the ten-thousands place in 49317.20568? _____</p> <p>(13) $64 \times 15 =$ _____</p> <p>(14) 83670.2874 rounded to the hundredths place is _____ decimal</p> <p>(15) What is the remainder for $700126 \div 9$? _____</p> <p>(16) The number of even whole numbers between 343 and 358 is _____</p> <p>(17) $1 \times 10^{-2} + 4 \times 10^{-4} =$ _____ (decimal)</p> <p>(18) $26 \times 11 - 11 \times 21 =$ _____</p> <p>(19) MMLIX = _____ (Arabic Numeral)</p> | <p>*(20) $625 \times 239 =$ _____</p> <p>(21) $18 - 6 \div 3 =$ _____</p> <p>(22) $11 + 15 + 19 + 23 =$ _____</p> <p>(23) 12 weeks = _____ days</p> <p>(24) Which is larger: 0.33 or $\frac{1}{3}$? _____</p> <p>(25) $\frac{21}{24} - \frac{5}{24} =$ _____</p> <p>(26) $28 \times 32 =$ _____</p> <p>(27) 68% = _____ common fraction</p> <p>(28) If 48 ♥ costs 44¢, then 36 ♥ cost _____ ¢</p> <p>(29) $24 \div \frac{3}{8} =$ _____</p> <p>*(30) $360 \times 667 =$ _____</p> <p>(31) 75 nickels = _____ quarters</p> <p>(32) The sum of the two largest primes less than 50 is _____</p> <p>(33) 32 ounces = _____ cups</p> <p>(34) $975 \div 25 =$ _____</p> <p>(35) $8\frac{1}{3}\%$ = _____ common fraction</p> <p>(36) The LCM of 18 and 36 is _____</p> <p>(37) $93 \times 94 =$ _____</p> |
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- (38) $\frac{7}{24} + \frac{11}{24} - \frac{9}{24} =$ _____
- (39) The ratio, in cents, of 5 dimes to 3 quarters is _____
- *(40) $161 \times 224 =$ _____
- (41) $4^3 =$ _____
- (42) $22^2 =$ _____
- (43) The volume of a cube with edge 3 is _____
- (44) The perimeter of an isosceles triangle with congruent sides 12 and other side 16 is _____
- (45) $2^5 - 3^3 =$ _____
- (46) $9\frac{1}{4} \times 7\frac{1}{4} =$ _____ (mixed number)
- (47) What is the volume of a rectangular box that is 2.25-ft by 8-ft by 11-ft? _____ cubic feet
- (48) $120 \times 12.5\% =$ _____
- (49) If $x = \frac{3}{4}$, then $15 - 8x =$ _____
- *(50) $24 \times 2024 + 2025 =$ _____
- (51) What is the number, **k**, in the sequence:
0, 1, 1, 2, 3, 5, 8, **k**, 21, 34, ...? _____
- (52) $12\frac{3}{5} - 6\frac{9}{10} =$ _____ (mixed number)
- (53) If the diameter of a circle is 48, then the area of the circle is _____ π
- (54) If a rectangle with area 32 has a length that measures 8, then its perimeter is _____
- (55) 46 (base 10) = _____ (base 5)
- (56) The number of days in December is _____
- (57) What is the area of a triangle with altitude $4\frac{1}{3}$ and base 12? _____
- (58) How many elements are in the power set for {A, U, S, T, I, N}? _____
- (59) If three times a whole number, added to six, is thirty-nine, what is the number? _____
- *(60) 1 mile = _____ inches
- (61) 1110 (base 2) = _____ (base 8)
- (62) $18 - (-2^3) \times (2) =$ _____
- (63) What is the probability a red marble is randomly picked from a bag of 16 blue and 8 red marbles? _____
- (64) 4.25 gallons = _____ pints
- (65) $82^2 =$ _____
- (66) $10^6 \div 8$ has remainder of _____
- (67) What is the perimeter of a rhombus with side 4.125? _____
- (68) If $6x - 14 > 58$, then $x >$ _____
- (69) $2\frac{4}{15} - \frac{5}{3} =$ _____
- *(70) $\sqrt{79524} =$ _____
- (71) The multiplicative inverse of $4\frac{3}{8}$ is _____
- (72) $16^2 - 24^2 =$ _____
- (73) If $6\frac{1}{2}\%$ of 12 is $2\frac{1}{6}\%$ of x , then $x =$ _____
- (74) $1^0 - 2^1 + 3^2 - 4^3 =$ _____
- (75) $8^2 + 24^2 =$ _____
- (76) $(1 + 3 + 5 + \dots + 11)^2 =$ _____
- (77) What is the area of a right triangle with hypotenuse 15 and leg 9? _____
- (78) $\sqrt{361} \times \sqrt{441} =$ _____
- (79) $29 + 29^2 =$ _____
- *(80) $15^4 =$ _____

2023 – 2024 University Interscholastic League Elementary Number Sense Test B – Key

(1) 48	*(20) 141907 – 156843	(38) $\frac{3}{8}$; .375	(59) 11
(2) 108	(21) 16		*(60) 60192 – 66528
(3) 62	(22) 68	(39) $\frac{2}{3}$	(61) 16
(4) 112	(23) 84		(62) 34
(5) 85	(24) $\frac{1}{3}$	*(40) 34261– 37867	(63) $\frac{1}{3}$
(6) 567		(41) 64	(64) 34
(7) 45	(25) $\frac{2}{3}$	(42) 484	(65) 6724
(8) 297	(26) 896	(43) 27	(66) 0
(9) 80		(44) 40	
*(10) 2853 – 3153	(27) $\frac{17}{25}$	(45) 5	(67) $16.5; 16\frac{1}{2}; \frac{33}{2}$
(11) 425	(28) 33	(46) $67\frac{1}{16}$	(68) 12
(12) 4	(29) 64	(47) 198	(69) $\frac{3}{5}$; .6
(13) 960	*(30) 228114– 252126	(48) 15	*(70) 268 – 296
(14) 83670.29	(31) 15	(49) 9	
(15) 7	(32) 90	*(50) 48071 – 53131	(71) $\frac{8}{35}$
(16) 7	(33) 4	(51) 13	(72) -320
(17) .0104	(34) 39	(52) $5\frac{7}{10}$	(73) 36
(18) 55	(35) $\frac{1}{12}$	(53) 576	(74) -56
(19) 2059	(36) 36	(54) 24	(75) 640
	(37) 8742	(55) 141	(76) 1296
		(56) 31	(77) 54
		(57) 26	(78) 399
		(58) 64	(79) 870
			*(80) 48094 – 53156

Note: *(Number) x – y means an integer between x and y inclusive.

If an answer is of the type like $\frac{2}{3}$ it cannot be written as .666... or $\overline{.6}$.